Self-correcting grounding: tiered perturbation and the nudge_crop correction loop

locate_tooth(FDI tooth)
returns a reference box (expert-verified box for annotated teeth; detector box otherwise)

Perturbation roll (fixed rates)
applied to the box SHOWN to the policy, never to the logged reference box

none - 45%
box unchanged
clean accept-and-proceed demonstration

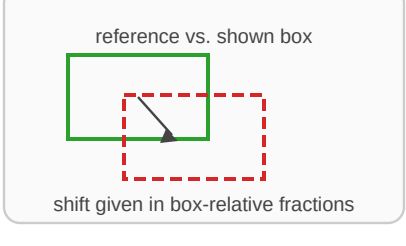
small - 25%
offset 12-28% of box size, random sign
genuine judgment call: often still fine given zoom padding - never resolved by the prompt

big - 30%
offset 45-75% of box size, random sign
visually self-evident without any verbal cue

Decoupled from the detector
rates are configured constants, not the detector's real error rate, so demonstrations stay valid as the grounding tool improves.

No verbal cue
an error is never announced in text; magnitude alone must carry the signal, so the learned behavior survives to GRPO and inference.

Teacher inspects the crop (zoom_crop)
and decides from the pixels alone whether the box is acceptable

nudge_crop
dx_frac, dy_frac, scale

zoom_crop on the corrected box - verify - commit finding
the same judgment the deployed policy faces at inference, with no ground truth available